Unit-6 - Instrumentation - Process control

Q.1. State the difference between feedback & feedforword controll system.

feed back control system	feed forword controll system
1) In feedback system output is depends on the generated feedback signals	1) In feed forword system the signal is pass to some external load.
2) measure of disturbance in the system is not needed by feedback system	2) measure of disturbance in the system is needed in feed forword system
3) All the disturbance are detected in the feedback system	3) All the disturbance are not detected in feed forword system
4) The loop in the feedback system is close loop	4) The loop in the feedforwork system is open loop.
5) It focous on the output of the system	5) It focous on the input of the system
6) The variable are adjusted on the basis of errors	6) The variable are adjusted on the basis of knowlege.
7) ~~No~~ stabliity is the most important	7) No stability problem
8) No accurate model need	8) Accurate model needed for both control & disturbance path.
9) Adaptable to non linear time variesd system	9) not adaptable for non linear time varied system
10) controll mu based on controll error	10) controll mu base on disturbance

Q.2. Describe the heat exchanger automatic control system with block diagram.

→ consider a counter current double pipe heat exchanger which cold fluid at tempreture T_{ci} is heated to temperature T_{co} by the hot fluid entering at higher temparature ~~Thi~~ Thi

The objective of controll system is to maintain temperature of cold fluid outlet stream (T_{oh}) inspite of the disturbance such as flow rates and temperature of hot and cold fluids entering the heat exchanger

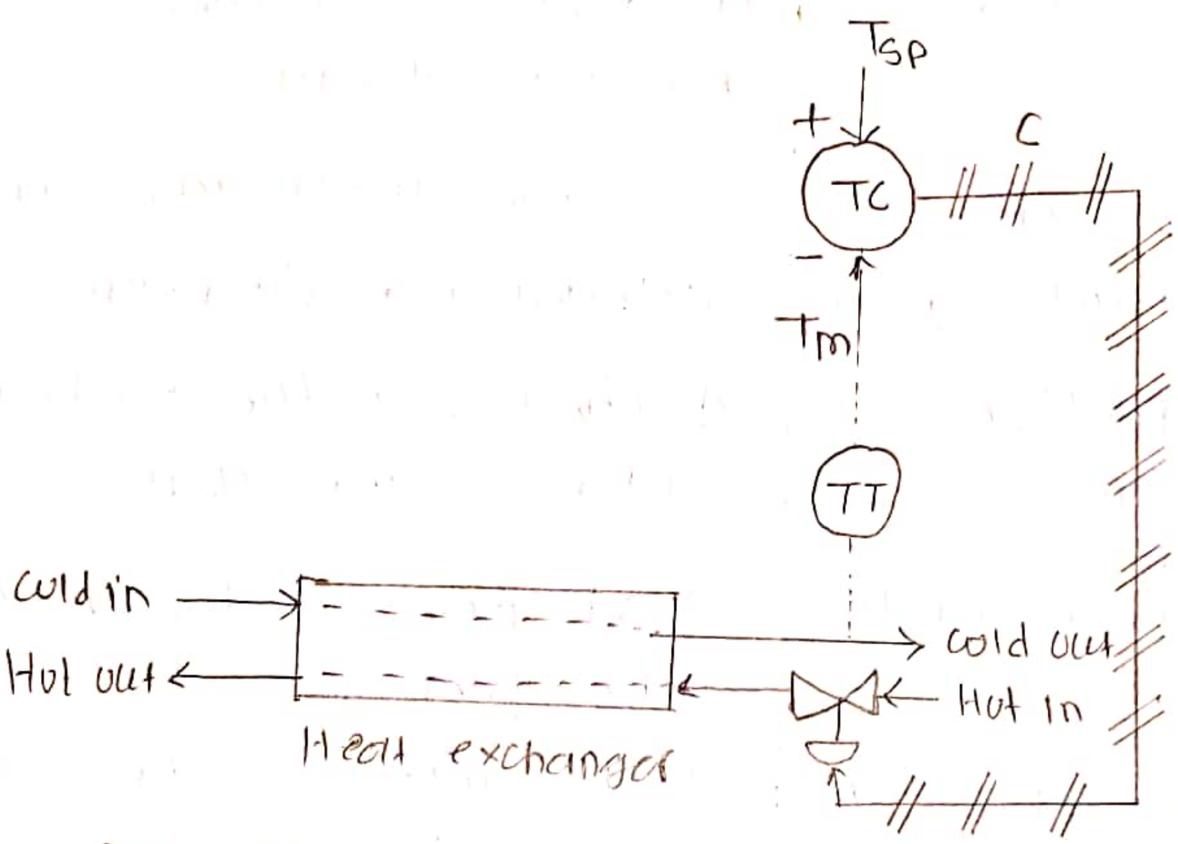


Fig - Temperature controll of heat exchanger.

- above figure shows controll strategy in which a suitable temperature sensor sense the temperature of the cold fluid outlet stream and temperature transmittor (TT) convert in into measurement signal (T_m).
- The Temperature controller (TC) compares the measurement signals with the set-point signal (T_{sp}) and calculate error signal $e = (T_{sp} - T_m)$.
- In response to this error signal, the controller generate the correction or command signal which opens or closes the controll valve installed in the pipe line of hot

fluid entering the heat exchanger, which in turn manipulate the temperature of cold fluid outlet stream at the desired set point value.

- In this feedback controlled strategy, controller action is based on the feedback from the temperature of cold fluid outlet stream.

Q.3 Derive transfer function of single tank liquid level system, response equation for the process if the input is given as a step change.

→

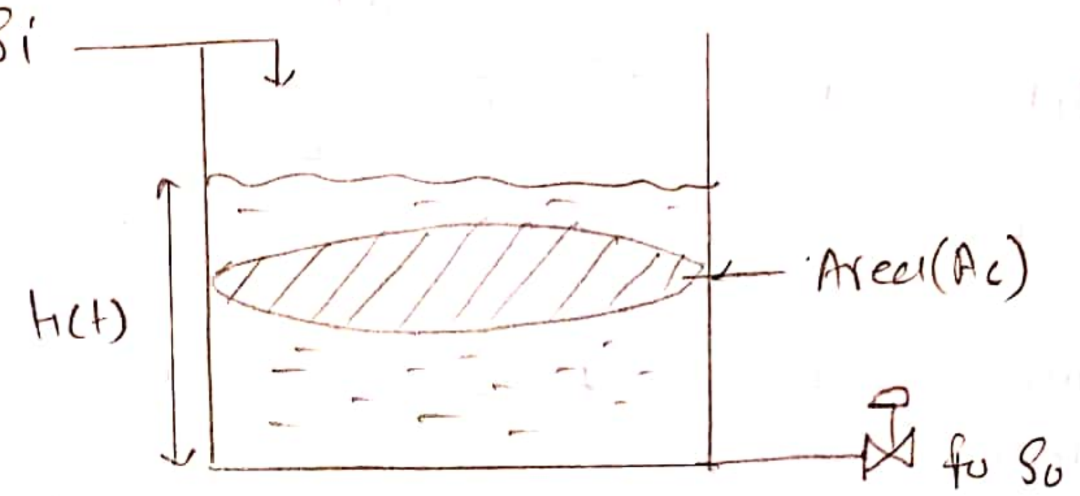


Fig liquid level system.

Above figure shows the liquid level system consisting of cylindrical tank having cross-section area A_c. A liquid of constant density ρ enters the tank at volumetric flow rate of f_i (volume/time). The resistance R is associated with pumps, valves, weirs & inlet or exit pipe lengths.

Here $H(t)$ = Height of liquid level $h = h(t)$

A_c = Cross sectional area of tank

$$\frac{\text{volume of tank}}{\text{height of tank}} \quad (A_c) = \frac{V}{h}$$

f_i = inlet flow rate in m^3/sec

f_o = outlet flow rate in m^3/sec

Assume $\rho_i = \rho_o = \rho$ fluid is incompressible.

$$f_o = \frac{h(t)}{R} \quad \text{—— (A)}$$

R = value resistance.

Apply mass balance at unsteady state

Rate of mass in − Rate of mass out = Accumulation

$$\rho_i f_i - \rho_o f_o = \frac{dm}{dt} \quad \text{—— (I)}$$

Now we know that $m = \rho \cdot V$

$$m = \rho \times A_c \times h$$

$$\frac{dm}{dt} = \rho \cdot A_c \times \frac{dh}{dt}$$

$$\cancel{\rho_i} f_i - \cancel{\rho_o} f_o = \cancel{dm} \; \cancel{\rho} A_c \cdot \frac{dh}{dt}$$

$$f_i - f_o = A_c \cdot \frac{dh}{dt} \quad \text{—— (2)}$$

At any time(t) $f_i(t) - f_o(t) = A_c \dfrac{dh(t)}{dt}$ —— (3)

At steady state :-

$$f_i(s) - f_o(s) = A_c \frac{dh(s)}{dt} \quad \text{—— (4)}$$

equation (3) − (4)

$$[f_i(t) - f_i(s)] - [f_o(t) - f_o(s)] = A_c \frac{d}{dt}[h(t) - h(s)]$$

According to defination of transfer function

$$F_i(t) - F_o(t) = A_c \frac{d}{dt}\{H(t)\}$$

$$F_i(t) - \frac{H(t)}{R} = A_c \frac{d}{dt}\{H(t)\} \quad \text{— from eqn (A)}$$

Taking laplace,

$$F_i(S) - \frac{H(S)}{R} = A_c\{S\,H(S) - \cancel{H(0)}\}$$

$$F_i(S) = H(S)\left\{\frac{1}{R} + A_c S\right\}$$

$$\frac{\text{outlet response}}{\text{Inlet flowrate}} : \frac{H(S)}{F_i(S)} = \frac{R}{A_c R S + 1}$$

Now.

the transfer function $\boxed{\dfrac{H(S)}{F_i(S)} = \dfrac{K_p}{\tau S + 1}}$ ...... $\tau = A_c \cdot R$

$G(S) = \uparrow$

Q4 Explain modes of controller with transfer function.

→ Following are the modes of controllers given below.

① P-controller.

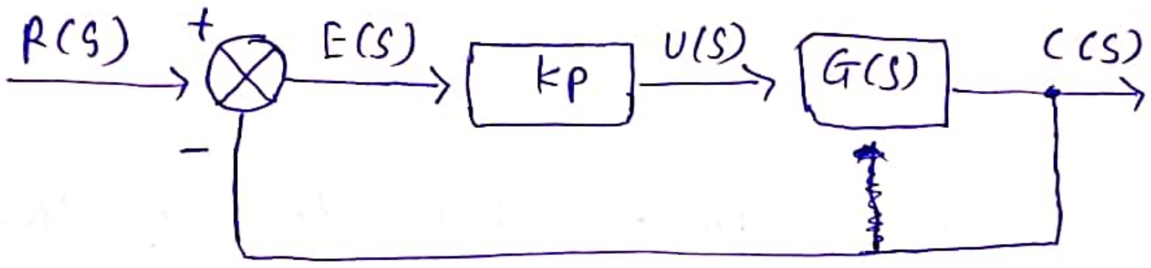
The proportional controller produces an output which is proportional to error signal.

$$u(t) = K_p e(t)$$

Apply laplace transform on both side

$$U(s) = K_p E(s)$$

$$\boxed{\frac{U(s)}{E(s)} = K_p}$$



The proportional controller is used to change the transient response as per the requirement.

② D-controller.

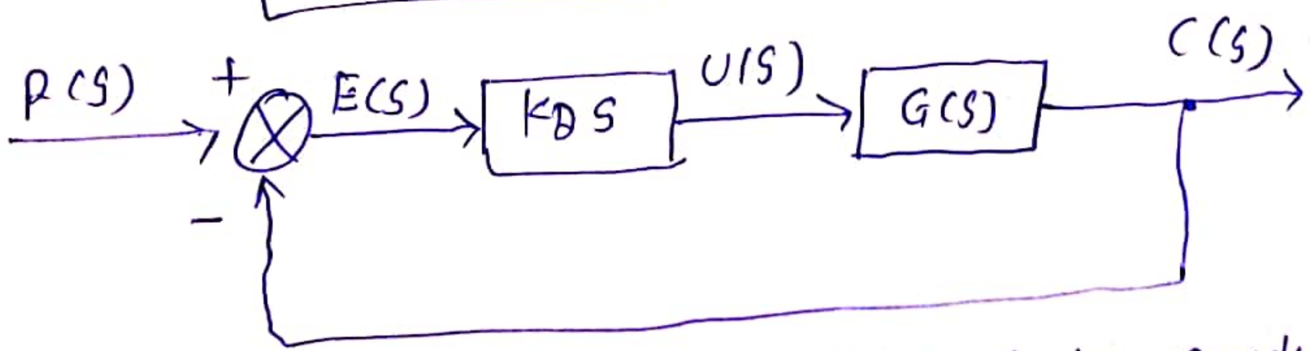
The Derivative controller produces an output which is derivative of the error signal.

$$u(t) = K_D \frac{de(t)}{dt}$$

Taking laplace transfor on both side.

$$U(s) = K_D s E(s)$$

$$\boxed{\frac{U(s)}{E(s)} = K_D s}$$



The derivative controller is used to ~~produces an output~~ ~~which is~~ ~~integral~~ make the unstable controll system into a stable one.

③ I- controller.

The integral controller produces an output which is integral of the error signal

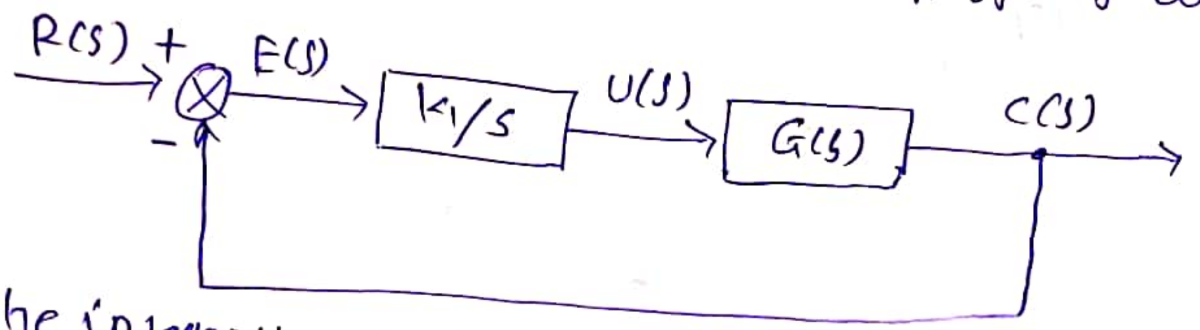
$$u(t) = k_I \int e(t)\,dt.$$

Taking laplace on both side.

$$U(s) = \frac{k_I E(s)}{s}$$

$$\boxed{\frac{U(s)}{E(s)} = \frac{k_I}{s}}$$

Therefor the transfer function of I-controller is $\left(\frac{k_I}{s}\right)$



The integral controller is ~~to~~ used to decrease the steady state errors.

④ PD- controller:

The proportional derivative controller produces an output which is the combination of output of proportional & derivative controllers.

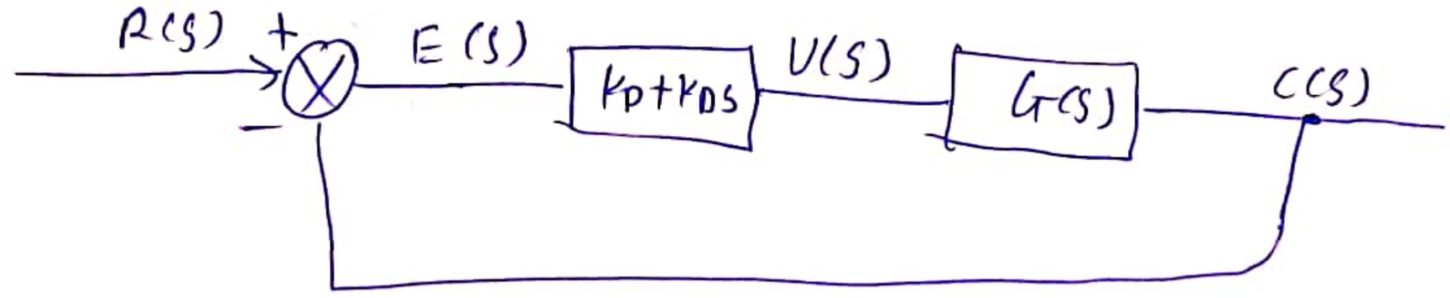
$$u(t) = k_p e(t) + k_D \frac{de(t)}{dt}$$

Taking laplace on both side.

$$U(s) = (k_p + k_D s)\,E(s)$$

$$\boxed{\frac{U(s)}{E(s)} = k_p + k_D s}$$

Therfore the transfer function of PD controller is $\underline{k_p + k_D s}$



The PD- controller is used to improve the stability of control system without affecting the steady state errors.

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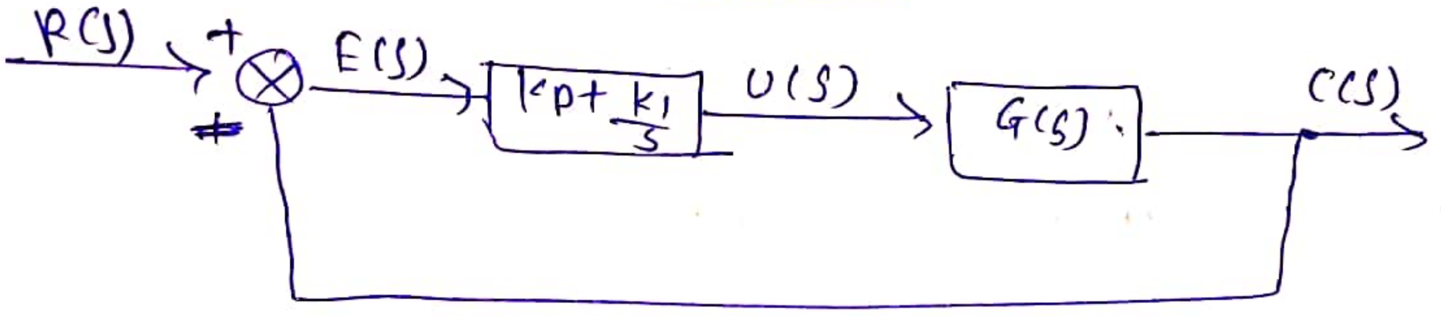
The proportional integral controller produces an output, which is the combination of outputs of the ~~PI~~ proportional & integral controllers.

$$u(t) = k_p e(t) + k_I \int e(t)\, dt$$

Taking laplace on both side

$$U(s) = \left(k_p + \frac{k_I}{s}\right) E(s)$$

$$\boxed{\frac{U(s)}{E(s)} = \left(k_p + \frac{k_I}{s}\right)}$$



The proportional integral controller is used to decrease the steady state error without affecting the stability of the control system.

(c) PID controller.

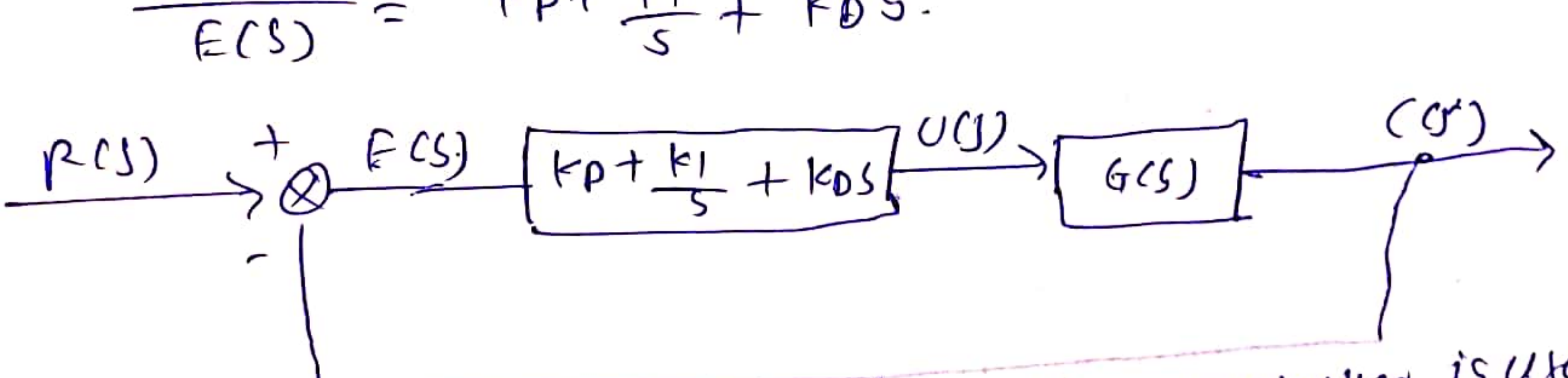
The proportional integral derivative controller produces an output, which is the combination of the outputs of proportional integral & derivative controllers.

$$u(t) = k_p e(t) + k_I \int e(t)\, dt + k_D \frac{d e(t)}{dt}$$

Taking laplace on both side

$$U(s) = \left(k_p + \frac{k_I}{s} + k_D s\right) E(s)$$

$$\frac{U(s)}{E(s)} = k_p + \frac{k_I}{s} + k_D s.$$



The proportional integral derivative controller is used to improve the stability of the control system and to decrease steady state errors.

Q.5 classification of variable in process control.

→ Process variable mainly classified in two different types.

- Input variable
- output variable.

① Input variable:

Input variable which denote the effect of surrounding to the chemical process.

Input variables further classified as.

1) manupulated (adjustable) variable

If there values can adjusted freely by the Human operator or controll mechanism.

2) Disturbances: If there values are not the result of adjustment by an operator or controll system.

② output variable:

output variables denotes the effect of process on the surrounding.

output variables are further classified as

1) measured output variable: If there value are known by directly measuring them.

eg. Temp., pressure, flow rate etc.

2) unmeasured output variables: If they are not or cannot be measured directly.